

Fubini-Tonelli:

$f: [a, b] \times [c, d] \rightarrow \mathbb{R}$, Riemannovo int.

Pisimo $f_x: [c, d] \rightarrow \mathbb{R}$, $f_x(y) = f(x, y)$

Če f_x int. na $[c, d]$ $\forall x \in [a, b]$, velja:

$$\iint_{[a, b] \times [c, d]} f(x, y) dx dy = \int_a^b \left(\int_c^d f(x, y) dy \right) dx.$$

→ Podobno tudi obratno (int. najprej po x in potem po y)

Fubini (osnovni izrek):

$f: [a, b] \times [c, d] \rightarrow \mathbb{R}$ zvezna $\Rightarrow$

$$\iint_{[a, b] \times [c, d]} f(x, y) dx dy = \int_a^b \left(\int_c^d f(x, y) dy \right) dx = \int_c^d \left(\int_a^b f(x, y) dx \right) dy.$$

20 $a, b > 0$. Izračunaj $\int_0^1 \frac{x^b - x^a}{\ln x} dx$. Ali je to osh. ali pospl. riemann. int.?

$$\int_0^1 \frac{x^b - x^a}{\ln x} dx = \int_0^1 \frac{x^y}{\ln x} \Big|_{y=a}^{y=b} dx =$$

$$= \int_0^1 \int_a^b x^y dy dx = \int_a^b \int_0^1 x^y dx dy = \int_a^b \left. \frac{x^{y+1}}{y+1} \right|_{x=0}^{x=1} dy =$$

$f(y) = x^y$ zvezna

$$= \int_a^b \frac{1}{y+1} dy = \ln|1+y| \Big|_a^b = \ln \frac{1+b}{1+a}$$

$$\lim_{x \rightarrow 1} \frac{x^b - x^a}{\ln x} \stackrel{\text{L'H.}}{=} \lim_{x \rightarrow 1} \frac{bx^{b-1} - ax^{a-1}}{1/x} = \underline{b-a}$$

$\Rightarrow$ imamo osh. riemann. int.

21. Pokaži, da rezultat velja tudi za $a, b > -1$.

Poskusimo z istim izračunom:

- predp., da $b > a$

$$\int_0^1 \frac{x^b - x^a}{\ln x} dx \stackrel{①}{=} \int_0^1 \int_a^b x^y dy dx \stackrel{②}{=} \int_a^b \int_0^1 x^y dx dy \stackrel{③}{=} \int_a^b \frac{1}{1+y} dy \stackrel{④}{=} \ln \frac{1+b}{1+a}$$

1) velja za $x \in (0, 1)$ ✓

2) velja: x^y je int. po x in po y . Poleg tega je $x^y > 0 \Rightarrow$ riemann. int. $\in [0, \infty]$.

3, 4) velja, ker $a, b > -1$

✓

če $f(x, y) \geq 0 \Rightarrow$ vrednost integrala je neodv. od vrstnega reda integracije - tudi, ko je integral enak ∞

spomnimo se:

$$\frac{d}{dx} \log \sqrt{x^2 + y^2} = \frac{1}{2} \frac{2x}{x^2 + y^2} = \frac{x}{x^2 + y^2}$$

$$\frac{d^2}{dx^2} \log \sqrt{x^2 + y^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

23. Izračunaj

$$\int_0^1 \int_0^1 \frac{y^2 - x^2}{(x^2 + y^2)^2} dx dy. \text{ Ali lahko zamenjamo vrstni red?}$$

$$\int_0^1 \int_0^1 \frac{y^2 - x^2}{(x^2 + y^2)^2} dx dy = \int_0^1 \frac{x}{x^2 + y^2} \Big|_0^1 dy = \arctan y \Big|_0^1 = \frac{\pi}{4}$$

$\rightarrow$ če zamenjamo vrstni red:

$$\int_0^1 \int_0^1 \frac{y^2 - x^2}{(x^2 + y^2)^2} dy dx = -\frac{\pi}{4} \Rightarrow \text{ne moremo zamenjati} - \text{problem: integrand postane neomejen pri izhodišču, če gremo čez diagonalo se spremeni predznak}$$

Eulerjeva Γ funkcija:

$$\Gamma(s) = \int_0^{\infty} t^{s-1} e^{-t} dt$$

Lastnosti:

$$- n \in \mathbb{N}: \Gamma(n+1) = n!$$

$$- \Gamma(s+1) = s \cdot \Gamma(s)$$

$$- \Gamma(1/2) = \sqrt{\pi}$$

$$- \Gamma(s) \Gamma(1-s) = \frac{\pi}{\sin(\pi s)}$$

Eulerjeva β funkcija:

$$\beta(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt = \frac{\Gamma(x) \Gamma(y)}{\Gamma(x+y)}$$

Lastnosti:

$$- \beta(x, y) = \beta(y, x)$$

$$- \beta(x, y) = \int_0^{\infty} \frac{u^{x+y-1}}{(1+u)^{x+y}} du$$

$$- \beta(x, y) = 2 \int_0^{\pi/2} \cos^{2x-1}(u) \sin^{2y-1}(u) du$$

izračunaj $\int_0^{\infty} x^4 e^{-x^2} dx$ $t = x^2$
 $dt = 2x dx$

$$\int_0^{\infty} x^4 e^{-x^2} dx = \frac{1}{2} \int_0^{\infty} t^2 e^{-t} dt = \Gamma(3) \cdot \frac{1}{2} = 2! \cdot \frac{1}{2} = \underline{\underline{2}}$$

$$\int_0^1 (\ln x)^4 \sqrt{-\ln x} dx = \int_0^1 t^{5/2} e^{-t} dt = \int_0^{\infty} t^{5/2} e^{-t} dt = \Gamma(7/2) = \frac{9}{2} \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi} = \frac{27 \cdot 35}{32} \sqrt{\pi}$$

$$-\ln x = t \rightarrow x = e^{-t}$$

$$-\frac{dx}{x} = dt$$

$$\int_{-1}^1 \sqrt{1-x^2} dx = 2 \int_0^1 \sqrt{1-x^2} dx = \int_0^1 u^{-1/2} (1-u)^{1/2} du = \beta\left(\frac{1}{2}, \frac{3}{2}\right) = \frac{\Gamma(1/2) \Gamma(3/2)}{\Gamma(2)} = \frac{\sqrt{\pi} \cdot \frac{\sqrt{\pi}}{2}}{1} = \frac{\pi}{2}$$

$$x^2 = u$$

$$2x dx = du$$

$$\int_0^4 x^2 \sqrt{16-x^2} dx = 4 \int_0^4 x^2 \sqrt{1-\left(\frac{x}{4}\right)^2} dx = 4 \int_0^1 24 \cdot \sqrt{u} \sqrt{1-u} du = 128 \beta\left(\frac{3}{2}, \frac{3}{2}\right) = 128 \frac{\left(\frac{1}{2}\right)^2 \sqrt{\pi}^2}{\Gamma(3)} =$$

$$= \frac{128}{4 \cdot 2} \pi = 16\pi$$

$$\left(\frac{x}{4}\right)^2 = u$$

$$\frac{1}{16} \cdot 2x dx = du$$